

BMW Market Analytics API Documentation

COMP3011 – Web Services and Web Data

1. Overview

This document describes all endpoints exposed by the BMW Market Analytics API.

Base URL (local development):

<http://127.0.0.1:8000>

All responses are returned in JSON format.

2. Authentication

Write operations require an API key.

Header:

X-API-Key:

Example:

X-API-Key: super-secret-key

If the key is missing or invalid:

Status Code: 401 Unauthorized

Response:

```
{  
  "detail": "Invalid or missing API key"  
}
```

Read-only endpoints do not require authentication.

3. Endpoints

3.1 Health

GET /health

Description:
Returns API status.

Authentication: Not required

Example Request:
GET /health

Example Response (200 OK):
{
 "status": "ok"
}

3.2 Car Models

GET /models

Description:
Returns car models. Supports optional filtering and pagination via query parameters.

Authentication: Not required

Query Parameters:

Parameter	Type	Default	Description
name	string	—	Filter by name (partial, case-insensitive)
series	string	—	Filter by series (partial, case-insensitive)
fuel_type	string	—	Filter by fuel type (exact match)
transmission	string	—	Filter by transmission (exact match)
skip	integer	0	Number of rows to skip (pagination)
limit	integer	100	Max rows to return (1-500)

Example Request:
GET /models?fuel_type=Diesel&limit=10

Example Response:

```
[
{
  "id": 1,
  "name": "3 Series",
  "series": "3 Series",
  "body_style": null,
  "fuel_type": "Petrol",
  "transmission": "Automatic"
}
```

GET /models/{id}

Path Parameter:

- id (integer)

Description:

Returns a single model by ID.

Authentication: Not required

Example Request:

GET /models/1

Example Response:

```
{
  "id": 1,
  "name": "3 Series",
  "series": "3 Series",
  "body_style": null,
  "fuel_type": "Petrol",
  "transmission": "Automatic"
}
```

Error (404 Not Found):

```
{
  "detail": "Model not found"
}
```

POST /models

Authentication: Required

Request Body:

```
{  
  "name": "X5",  
  "series": "X5",  
  "body_style": "SUV",  
  "fuel_type": "Diesel",  
  "transmission": "Automatic"  
}
```

Example Request:

POST /models

Example Response (201 Created):

```
{  
  "id": 10,  
  "name": "X5",  
  "series": "X5",  
  "body_style": "SUV",  
  "fuel_type": "Diesel",  
  "transmission": "Automatic"  
}
```

PATCH /models/{id}

Authentication: Required

Path Parameter:

- id(integer)

Request Body Example:

```
{  
  "fuel_type": "Hybrid"  
}
```

Example Request:

PATCH /models/1

Example Response:

```
{  
  "id": 1,  
  "name": "3 Series",
```

```
"series": "3 Series",
"body_style": null,
"fuel_type": "Hybrid",
"transmission": "Automatic"
}
```

DELETE /models/{id}

Authentication: Required

Example Request:

```
DELETE /models/1
```

Success Response:

Status Code: 204 No Content

3.3 Market Records

GET /models/{id}/records

Path Parameter:

- id (integer)

Description:

Returns price records for a model. Supports optional filtering and pagination via query parameters.

Authentication: Not required

Query Parameters:

Parameter	Type	Default	Description
year	integer	—	Filter by exact year (1900–2100)
min_price	float	—	Minimum price, inclusive (> 0)
max_price	float	—	Maximum price, inclusive (> 0)
skip	integer	0	Number of rows to skip (pagination)
limit	integer	200	Max rows to return (1–1000)

Example Request:

GET /models/1/records?year=2019&min_price=10000

Example Response:

```
[
{
  "id": 20,
  "car_model_id": 1,
  "year": 2018,
  "price": 19000.0,
  "sales_volume": null
}
```

POST /models/{id}/records

Authentication: Required

Path Parameter:

- id(integer)

Request Body:

```
{
  "year": 2018,
  "price": 19000
}
```

Example Request:

POST /models/1/records

Example Response (201 Created):

```
{
  "id": 25,
  "car_model_id": 1,
  "year": 2018,
  "price": 19000.0,
  "sales_volume": null
}
```

Error (404 Not Found):

```
{
```

```
"detail": "Model not found"
}
```

GET /models/{id}/records/{record_id}

Path Parameters:

- `id` (integer) — car model ID
- `record_id` (integer) — market record ID

Description:

Returns a single market record by ID. Returns 404 if the record does not exist or does not belong to the specified model.

Authentication: Not required

Example Request:

```
GET /models/1/records/20
```

Example Response (200 OK):

```
{
  "id": 20,
  "car_model_id": 1,
  "year": 2018,
  "price": 19000.0,
  "sales_volume": null
}
```

Error (404 Not Found):

```
{
  "detail": "Record not found"
}
```

PATCH /models/{id}/records/{record_id}

Path Parameters:

- `id` (integer) — car model ID
- `record_id` (integer) — market record ID

Authentication: Required

Description:

Partially updates a market record. Only fields included in the request body are modified; omitted fields are left unchanged.

Request Body Example:

```
{  
  "price": 17500.0  
}
```

Example Request:

PATCH /models/1/records/20

Example Response (200 OK):

```
{  
  "id": 20,  
  "car_model_id": 1,  
  "year": 2018,  
  "price": 17500.0,  
  "sales_volume": null  
}
```

Error (404 Not Found):

```
{  
  "detail": "Record not found"  
}
```

DELETE /models/{id}/records/{record_id}

Path Parameters:

- id (integer) – car model ID
- record_id (integer) – market record ID

Authentication: Required

Example Request:

DELETE /models/1/records/20

Success Response:

Status Code: 204 No Content

Error (404 Not Found):

```
{
```



```
"detail": "Record not found"
```

```
}
```

3.4 Analytics Endpoints

GET /analytics/average-price

Query Parameter:

- `model_id` (integer, required)

Example Request:

```
GET /analytics/average-price?model_id=1
```

Example Response:

```
{  
  "model_id": 1,  
  "model_name": "3 Series",  
  "average_price": 18250.75  
}
```

GET /analytics/price-trend

Query Parameter:

- `model_id` (integer, required)

Example Request:

```
GET /analytics/price-trend?model_id=1
```

Example Response:

```
{  
  "model_id": 1,  
  "model_name": "3 Series",  
  "trend": [  
    {  
      "year": 2016,  
      "average_price": 18000.0,  
      "num_records": 12  
    }  
  ]  
}
```

```
]
}
```

GET /analytics/top-models

Query Parameters:

- year (integer, required)
- limit (integer, optional)

Example Request:

GET /analytics/top-models?year=2018&limit=5

Example Response:

```
{
  "year": 2018,
  "results": [
    {
      "model_id": 3,
      "model_name": "X5",
      "average_price": 35000.0,
      "num_records": 7
    }
  ]
}
```

4. Error Codes Summary

Code	Meaning
200	Successful request
201	Resource created
204	Resource deleted
401	Invalid or missing API key
404	Resource not found
422	Validation error (invalid or missing field values)

5. Setup Notes

- Run database migrations before starting the API.
- Import dataset using: `python -m scripts.import_bmw`
- Start server with: `uvicorn app.main:app --reload`